Tut Test 6 [25 marks]

Consider the system with the transfer function $G(s) = \frac{18}{(s+2)(s+3)}$

This system is to be placed in negative feedback with a lead compensator to satisfy the constraints:

i. Settling time decreased to 50% compared to the open-loop system.
ii. Damping ratio of at least 0.6.
iii. Damped frequency of oscillation of less than 10 rad/s
iv. Accuracy with respect to position setpoints of more than 90%

a) Draw the root locus of the system, and indicate the boundaries within which the transient response specifications i-iii will be satisfied, and hence, confirm that a complex pole pair at $-5 \pm 5j$ meets these requirements. [8 marks]

b) Why is it not possible to satisfy the specifications with proportional control? [2 marks]

c) Design a lead compensator for the system that places the dominant closed-loop poles at the suggested location ($-5 \pm 5j$). State whether your design also satisfies specification iv, and if not, give a brief suggestion of how you can change your design to achieve greater accuracy. [15 marks]

Tut Test 6

Name: ____________________

Answer sheet

Student No: ____________________

a)

b)

c)